



Statistical Properties and Community Detection Study of Modern Chinese Synonym Networks

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Abstract. This paper constructs a synonym network using the vector representations of 53,946 words listed in the *List of Commonly Used Modern Chinese Words*, with words as nodes and synonym relations as edges, resulting in a network with 26,485 nodes and 71,964 edges. The findings indicate that the network exhibits scale-free, small-world properties. Through community detection, the network can be decomposed into several sub-networks, each representing a distinct synonymous semantic field. Consequently, this study may shed light on the complexity of lexicon networks and the modernization of dictionary compilation.

Keywords: word vector · lexicon network · synonym network · scale-free · small-world · dictionary compilation

1 Introduction

Saussure [1] explicitly introduced the concept of “language as a system of signs”, which has garnered widespread acceptance and significant attention [2]. The systematization of language has been examined from multiple perspectives, ranging from symbolic systems to human-driven complex adaptive systems [3]. Hjelmslev [4, 5] examined language systematization through the lens of syntagmatic and paradigmatic relations. Tesnière [6] proposed dependency syntax, which connects linguistic units through dependency relations, thereby facilitating the study of syntactic systematization.

However, these studies often provide only a partial view of the language system due to limited materials. Some people [7, 8] argue that language functions as a complex system, where complex networks are vital tools for quantitatively investigating linguistic structures. Complex networks can be constructed to achieve quantitative analysis of language systems by utilizing the relationships between language symbols. For instance, syntactic dependency relationships can be used to create syntactic dependency networks [9–11], co-occurrence relationships between words can form co-occurrence networks [12–14], and semantic connections among words can lead to semantic networks [15–18].

In contrast to syntactic dependency and co-occurrence, the meanings of words are inherently fuzzy and challenging to formalize. Previous studies on lexicon networks have primarily relied on static resources. For instance, Muller et al. [19] used French dictionaries to assume that words with similar definitions are synonyms, constructing

synonym relations based on shared terms within definitions. Similarly, Makaruk and Owczarek [20] examined the Polish synonym network, arguing that it exhibits scale-free properties. Li et al. [18] focused on Chinese lexicons, demonstrating that the Chinese synonym network also displays scale-free properties and a hierarchical structure.

Unlike syntax, which tends to be more stable, words can change rapidly. A significant limitation of constructing lexicon networks in this manner is that static resources do not adapt quickly to semantic shifts, which may lead to a lag. For example, the meanings of words can change due to the emergence of new concepts or shifts in societal attitudes. Due to constraints such as the costs of time and resources, dictionaries often struggle to keep up with these changes on time.

This study employs distributional semantics, utilizing word vectors derived from large-scale corpus data. Focusing on the *List of Commonly Used Modern Chinese Words* [21], we assess the cosine similarity between word vectors to determine whether two words are synonymous. The *List of Commonly Used Modern Chinese Words* is developed based on a corpus of approximately 250 million characters, including 56,008 commonly used terms in modern Chinese. It comprises 3,181 monosyllabic words, 40,351 disyllabic words, 6,459 trisyllabic words, 5,855 tetrasyllabic words, and 162 words with five or more syllables. Using word vectors from the Python package *Synonyms* (<https://github.com/chatopera/Synonyms>), we calculate the cosine similarity among words from the *List of Commonly Used Modern Chinese Words*. Similarities between words greater than 0.9 are established as edges in our network. With the above efforts, we aim to address the following two questions:

- (1) what are the statistical properties of the Chinese synonym network?
- (2) how can the synonymous network be utilized to compile synonym dictionaries? Specifically, how can it facilitate the identification of semantic fields?

A semantic field denotes a segment of reality symbolized by a set of related words. The words in a semantic field share a common semantic property [22]. Compiling synonym dictionaries requires the categorization of different words into various semantic fields. When conceptualized, a semantic field can be represented as a complex network, where words are depicted as nodes, and connections between these nodes are established based on shared semantic features. Extracting semantic fields from the synonym network can enhance the practical applications of research in synonym networks.

2 Basic Concepts of the Complex Networks

To begin, we should introduce some basic concepts of complex networks. A complex network can be formalized as a graph $G = (V, E)$, where V denotes a set of nodes and E represents a set of edges. Each edge in E corresponds to a pair of nodes in V . Before analysing, it is important to define the main parameters we will use, which include:

Degree (k): The degree of a node refers to the number of neighbours it has. The degree distribution $P(k)$ represents the probability that a randomly selected node has a degree of k . If the network contains a total of n nodes, and n_k denotes the number of nodes that have a degree of k , the degree distribution can be expressed as Eq. (1).

$$P(k) = \frac{n_k}{n} \quad (1)$$

This formulation allows us to analyze the distribution of connectivity within the network.

If the degree distribution of a network follows a power-law distribution with a degree exponent between 2 and 3, the network is scale-free. This means that there are relatively few nodes with very high degrees and many nodes with very low degrees. Real-world networks typically possess this property.

Average Path Length (L): The average path length quantifies the average shortest path between any two nodes in the network. It reflects the closeness of connections among nodes and provides insight into the overall size of the network. A shorter average path length suggests a more interconnected network, facilitating quicker information flow between nodes.

$$l_G = \frac{1}{n(n-1)} \sum_{i \neq j} d(v_i, v_j) \quad (2)$$

The total number of node pairs in a network is $n(n-1)$, and $d(v_i, v_j)$ denotes the shortest path between node i and node j .

Network Diameter (D): The network diameter is defined as the longest of all the shortest paths between any two nodes in a network. In other words, it represents the maximum shortest path length.

Average Clustering Coefficient (C): The average clustering coefficient quantifies the tendency of nodes to cluster together in the network. It is calculated as the average of the local clustering coefficients for all nodes. The formula for the average clustering coefficient is given by:

$$C = \frac{1}{n} \sum_{i=1}^n C_i \quad (3)$$

where C_i is the local clustering coefficient of node i . This coefficient provides insight into the degree to which nodes in the network tend to form tightly-knit groups.

Networks displaying small-world properties typically satisfy the conditions $L \approx L_{\text{random}}$, $C \gg C_{\text{random}}$. The small-world property indicates the presence of central nodes within the network that enhance communication efficiency.

In the following sections, these metrics will be used to analyze the Chinese synonym network.

3 Network Analysis of the Chinese Synonym Network

This study extracts words from the *List of Commonly Used Modern Chinese Words* to calculate the cosine similarity between each pair of word vectors. An edge is established between two words if their cosine similarity exceeds 0.9, resulting in a network with 28,475 nodes and 73,291 edges.

However, this network is not fully connected. Using *Gephi*, we identified 771 connected components within our network. With *NetworkX*, we exported each connected component as an independent network. In this section, we focus on the largest connected

Table 1. Main Parameters of the Synonym Network.

Main Parameters	E	N	$\langle k \rangle$	D	C	L
Quantity	71964	26485	5.434	22	0.245	7.212

E : edge; N : nodes; $\langle k \rangle$: average degree; D : network diameter; C : average clustering coefficient; L : average path length.

component, which consists of 26,485 nodes. Additionally, there are 26 networks with nodes ranging from 5 to 16, while the remaining 744 networks consist of only two nodes.

Then, we will further investigate these parameters of the synonym network.

3.1 Degree Distribution of the Synonym Network

If the degree distribution follows power-law, it indicates that the network is scale-free. We fitted the degree distribution of our network to Eq. (1), resulting in the fitted curve shown in Fig. 1.

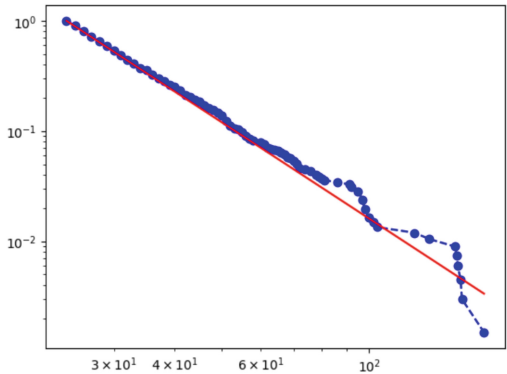


Fig. 1. The degree distribution of the synonym lexicon network.

In Fig. 1, the degree exponent γ is 3.194, with $R^2 = 0.944$. Typically, the degree exponent of a scale-free network will be within the range of 2 to 3 [23]. In our result, the relatively high value of γ suggests a weaker scale-free property. However, as shown in Table 1, the average clustering coefficient is 0.245, indicating that there is still a centralized structure in the network. Thus, the network can be seen as a scale-free network with some key nodes.

3.2 Small-World Properties of the Synonym Network

Watts and Strogatz [24] introduced the concept of small-world properties in social networks, which Solé et al. [25] identified as a statistically universal phenomenon. Furthermore, Steyvers and Tenenbaum [26] analyzed the structures of three semantic networks—word associations, WordNet, and Roget’s Thesaurus—and found that they all

exhibit small-world properties, characterized by short average path lengths between words, and strong local clustering.

To investigate this property, we generated 100 random networks with 26,485 nodes and 71,964 edges, calculating their average values for average path lengths and clustering coefficients. The results are as follows:

Average Path Length (L_{random}): 6.2073.

Clustering Coefficient (C_{random}): 0.00019628.

Comparing these values to the synonym network:

Average Path Length (L): 7.212.

Clustering Coefficient (C): 0.245.

We observe that $L \approx L_{random}$, $C \gg C_{random}$. Therefore, we conclude that the synonym network exhibits small-world characteristics.

3.3 Centrality Analysis of the Synonym Network

To assess the importance of nodes within the synonym network, we compute three centrality parameters: degree centrality, closeness centrality, and betweenness centrality.

Degree Centrality: Degree centrality reflects the extent to which a node is connected to all other nodes in the network. For an undirected graph, the degree centrality of node i can be expressed as:

$$C_D(i) = \frac{deg(i)}{n - 1} \quad (4)$$

where $deg(i)$ represents the degree of node i and n is the total number of nodes in the network. We calculate the degree centrality for each node in the synonym network and present the distribution in Fig. 2.

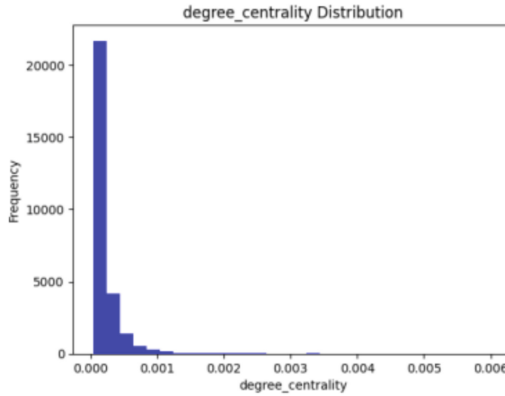


Fig. 2. The degree centrality distribution of the synonym network.

In Fig. 2, it is evident that the degree centrality of the vast majority of nodes is extremely low, with only a small number of nodes exhibiting an extremely higher degree centrality. This phenomenon is a further manifestation of the degree distribution.

Closeness Centrality: Closeness centrality, as defined by Newman [27], quantifies the proximity or accessibility of a node to all other nodes in the network. According to Caldarelli [28], the closeness centrality $C(i)$ of node i is defined as the reciprocal of the average shortest path length from i to all other nodes:

$$C(i) = \frac{1}{\sum_{j=1, n} d_{ij}} \quad (5)$$

where n represents the set of nodes reachable from node i , and d_{ij} denotes the distance between nodes i and j . We compute the closeness centrality for each node in the synonym network and present the results in Fig. 3.

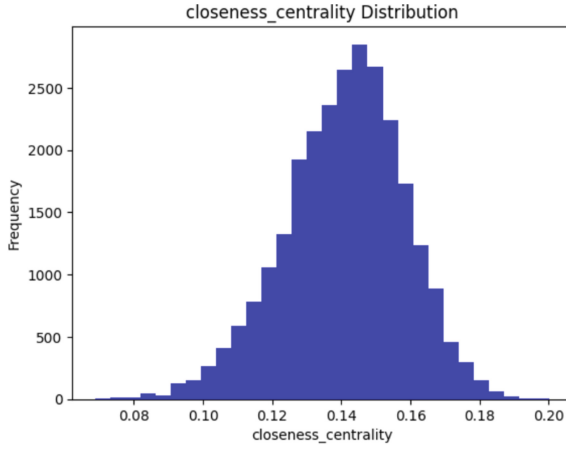


Fig. 3. The closeness centrality distribution of the synonym network.

A higher closeness centrality indicates that a node can more rapidly access other nodes within the network. The graph shows that the closeness centrality of nodes in the synonym network follows a Poisson distribution. Most nodes exhibit relatively concentrated closeness centrality values, with only a few nodes exhibiting exceptionally high or low values. This observation suggests that there are no words that share an extremely high closeness centrality.

Betweenness Centrality: Betweenness centrality measures the importance of a node as an intermediary within a network. A node is considered to have high betweenness centrality if it frequently appears on the shortest paths between other nodes. The betweenness centrality of node i , denoted as $B_C(i)$, can be calculated using the following formula [28]:

$$B_C(i) = \sum_{j \neq k} \frac{b_{jk}(i)}{b_{jk}} \quad (6)$$

where b_{jk} represents the total number of shortest paths between nodes j and k , and $b_{jk}(i)$ represents the number of those shortest paths that include node i . We compute the betweenness centrality for each node in the network, as illustrated in Fig. 4.

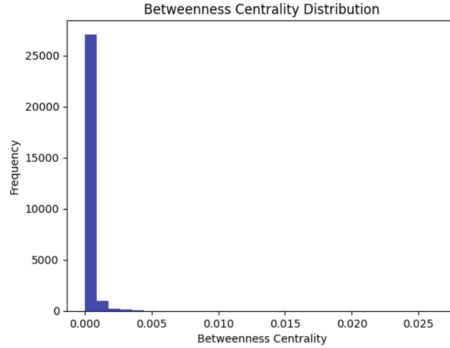


Fig. 4. Betweenness Centrality Distribution.

The results in Fig. 4 reveal that a disproportionately high percentage of nodes exhibit low betweenness centrality, while only a few nodes possess slightly higher values.

We computed the degree centrality, closeness centrality, and betweenness centrality for all nodes in the synonym network. The above results indicate that, despite a lack of significant differences in closeness centrality among nodes, there is a notable differentiation in degree centrality and betweenness centrality. This suggests that central nodes do exist within the network, although they may not be as prominent as those in syntactic and co-occurrence networks. Additionally, we calculated the degree centralization of the network, which reflects the relative strength of central nodes within the network [29]. For the synonym network, the degree centralization was computed to be 0.00586.

Combining the centrality values of nodes with the degree centralization of the network, we argue that synonym networks are not prototypically centralized, implying that not all linguistic networks exhibit a centralized structure. Chen and Liu [30] constructed syntactic dependency networks based on two genres of Chinese treebanks. They selected three Chinese function words as research objects and found that in the syntactic network, the function word 的(de, of) has the highest betweenness centrality and is the most central node in the entire network. Removing this central node resulted in some nodes becoming completely isolated, demonstrating characteristics of a centralized network, where the absence of a central node reduces network connectivity and increases average path lengths. The emergence of central nodes in syntactic networks is attributed to the reliance on function words in languages like Chinese. Without these words, it may become difficult to adhere to the principle of the least effort. However, the relatively weaker centrality in synonym networks suggests that indispensable concepts in semantics are less prevalent compared to function words. Findings from neurolinguistics also lend support, indicating that even when certain brain regions are damaged, patients rarely lose their entire vocabulary but may experience difficulties in word retrieval [31].

The analysis of the network structure suggests that multiple dense community structures may exist within the Chinese synonym network. In the following section, we will identify and analyze these communities, exploring their implications for dictionary compilation.

4 Community Detection in the Chinese Synonym Network

The compilation of synonym dictionaries represents a key application of research on synonym networks. Traditionally, lexicography involves summarizing scattered words into semantic fields, a process that is often time-consuming and labor-intensive, and can be influenced by subjective language intuition. We aim to identify semantic fields within the synonym network by leveraging complex network analysis. Specifically, we seek to discover communities within the complex network and assess whether these communities can be defined as semantic fields. This approach promises a more systematic and efficient method for compiling synonym dictionaries, enhancing both their accuracy and usability in linguistic applications.

4.1 Communities as Independent Components

We computed the cosine similarity among all words in the *List of Commonly Used Modern Chinese Words* and constructed a network. Within this network, we identified 771 independent connected components. In Sect. 3, our analysis focuses on the largest connected components, and other connected components are ignored. In these connected components, 26 share node counts ranging from 5 to 16. We present the subgraph corresponding to 16 nodes in Fig. 5.

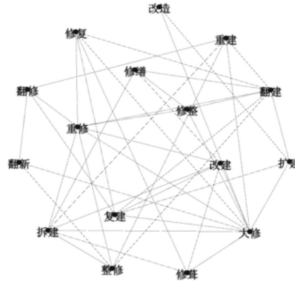


Fig. 5. Subgraph with 16 Nodes.

These nodes are: 修葺 (xiuqi, repair), 大修 (daxiu, major repair), 重建 (chongjian, reconstruction), 修缮 (xiushan, repair), 改造 (gaizao, remold), 重修 (chongxiu, rebuild), 修复 (xiufu, restore), 修整 (xiuzheng, adjust), 复建 (fujian, reconstruct), 整修 (zhengxiu, overhaul), 翻修 (fanxiu, renovate), 翻建 (fanjian, rebuild), 改建 (gaijian, remodel), 拆建 (chaijian, demolition), 扩建 (kuojian, expand), 翻新 (fanxin, refurbish).

By analyzing the meanings of these nodes in Fig. 5, we find that the words can be categorized into a semantic field related to building maintenance. Upon calculating the degree centrality, betweenness centrality, and closeness centrality for all nodes within this semantic field, we identify that the top three nodes in terms of these centralities are “major repair (大修),” “renovation (翻建),” and “reconstruction (重修).” This result indicates that non-largest connected components within synonym networks may be treated as semantic fields, providing a convenient approach to dictionary compilation.

4.2 Community Detection Within the Largest Connected Components

Semantic fields appear in non-largest connected components, which raises the question: do semantic fields exist in the largest one? In this section, we utilize the Louvain algorithm, developed by Blondel et al. [32], which effectively uncovers community structures in the largest connected component.

Applying the Louvain algorithm, we partitioned the synonym network, consisting of 26,485 nodes, into 66 subgraphs. The largest community contains 1,298 nodes, while the smallest has 14 nodes. Larger communities may be subdivided so that smaller networks can be derived through iterative clustering.

To examine the relationship between network scale—quantified by node count—and various parameters, we conduct a Spearman’s rho correlation analysis. The result can help us to judge when communities do not require a further split.

The node clustering aims to reveal potential communities, with nodes sharing closely related meanings. Thus, these small-scale communities can also be interpreted as semantic fields in the construction of a synonym dictionary. It is essential to determine the threshold for defining a small community. To this end, we will analyze the correlation between community scale (in terms of nodes and edges) and other network parameters. The results for the 66 subgraphs are presented below (Table 2):

Table 2. The Spearman’s rho correlation analysis of parameters of 66 subnetworks.

		$\langle k \rangle$	D	Density	Modularity	C
E	r	0.815	0.164	−0.854	0.277	0.138
	p	0	0.188	0	0.024	0.267
N	r	0.68	0.232	−0.943	0.402	0.024
	p	0	0.061	0	0.001	0.846

The correlation coefficient quantifies the degree of correlation between two variables, ranging from −1 to 1. A value close to 1 indicates a strong positive correlation, while a value near −1 reflects a strong negative correlation. A value of 0 signifies no correlation.

It shows that the number of edges is strongly positively correlated with the average degree ($p < 0.05$) and strongly negatively correlated with graph density ($p < 0.05$). The number of nodes is positively correlated with the average degree ($p < 0.05$) and strongly negatively correlated with graph density ($p < 0.05$). However, the positive correlation between the number of nodes and the average degree is weaker than the correlation between the number of edges and the average degree.

These findings suggest that as the number of edges increases, the average degree of the network also rises, while the graph deity decreases. This indicates that larger networks tend to have more connections per node, but are less densely connected overall. Understanding these relationships will help in determining appropriate thresholds for identifying semantic fields within the largest connected components.

In other words, as the network size gradually decreases, the average degree also decreases, while the graph density increases. This suggests that although the total number

of nodes in the network diminishes, the connections among nodes within communities become tighter, leading to a more compact and highly clustered structure.

By analyzing the semantics of words within small-scale subgraphs, we observe that networks with 33 or fewer nodes tend to share a common semantic field. Specifically, for subgraphs with node counts of 33 or less, the average degree is 2.602, and the average graph density is 0.1205. This reinforces the idea that smaller communities are more semantically cohesive, making them suitable for identification as distinct semantic fields in dictionary compilation.

By iteratively applying the Louvain algorithm, we generate numerous subgraphs. Spearman's rho correlation analysis indicates that as the network scale decreases, a more compact and highly clustered structure emerges, as evidenced by the changes in average degree and graph density.

Consequently, we propose using the number of nodes, average degree, and graph density as criteria to determine whether further clustering is necessary. Specifically, clustering can stop when the number of nodes is ≤ 33 , the average degree is approximately 2.602, and the graph density is around 0.1205.

In summary, the continuous clustering of large synonym networks can be effectively guided by these parameter values, facilitating the identification of meaningful semantic fields. This approach enhances the efficiency of compiling synonym dictionaries by ensuring that only relevant and closely related terms are grouped.

5 Conclusion

Returning to the two questions initially posed in this paper, we address them as follows:

The synonym network exhibits two key properties:

1. **Scale-Free property:** The degree distribution of the synonym network follows a power-law distribution, indicating a scale-free property. However, this property is relatively weak when compared with other linguistic networks. This may be due to more pronounced modularity within the conceptual space, where fewer concepts are indispensable, leading to a lesser presence of central nodes.
2. **Small-World Property:** The synonym network demonstrates small-world properties, suggesting that most nodes can be reached from any other node through a small number of connections.

Synonym networks can be clustered into smaller subgraphs by detecting communities within larger networks. Through parameter analysis, we propose that the number of nodes, average degree, and graph density can serve as criteria for determining the necessity of further clustering. When these parameters approach certain values, the words within subgraphs semantically align with a single semantic field. This insight can be directly applied to the compilation of synonym dictionaries, facilitating the organization of related terms into coherent entries.

In conclusion, our findings provide a framework for utilizing synonym networks in lexicography, enhancing both the efficiency and accuracy of synonym identification and classification.

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